INVESTIGATING TRENDS IN CRYPTO

Case study: Coinbase

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CONTENT

OUR TEAM

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WHY CRYPTO?

Crypto is uniquely **sensitive** to public sentiment, making it a real-time case study of **volatility**. A tweet, headline, or Reddit thread can impact a stock.

Understanding these dynamics can help develop faster, **sentiment-drive investment strategies.**

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Coinbase

- Largest cryptocurrency exchange in the U.S.
- $COIN reflects public trust and sentiment toward crypto markets

Recent Events

- Spot Bitcoin ETFs launched causing 40% Bitcoin surge (2024)
- Stricter crypto exchange causing COIN to fall 30% (Q1 2025)

Why Is It Relevant?

- Opportunity to capitalize on momentum, exploiting short-term strategies
- Sentiment monitoring (social media, news) becomes essential & can give an edge

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OUR GOAL

Our project aims to investigate how public sentiment data can inform crypto stock forecasting – using Coinbase ($COIN) as a case study.

We pursued three core objectives:

1. Analyze sentiment signals from social media (Reddit, X/Twitter), news headlines, and Google Trends.
2. Understand how these signals correlate with COIN's stock price, volatility, and event-driven movement.
3. Predict future price movements using machine learning models trained on both sentiment and historical financial data.

OVERVIEW

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Project Timeline

Nov 2024
Teams assigned

Jan 2025
Topic selection & research

Feb 2025
Data collection & API integration

Mar 2025
Sentiment scoring & modeling

Apr 2025
Forecasting & presentation building

01 Data Collection

- **Yahoo Finance** (COIN price, volume, volatility)
- **Google Trends** (search interest for “Coinbase” & “cryptocurrency”)
- **Reddit & Twitter** (posts mentioning Coinbase or $COIN)
- **News headlines** (via Google News API)

02 Data Preprocessing

- Data Cleaning
- Calculated financial indicators

03 Sentiment Analysis

- **VADER** (score sentiment for Reddit / Twitter / news headlines)
- **LDA topic modeling** + **BERTopic** (Reddit: identify recurring themes)
- Switched to **RoBERTa** after **VADER** limitations

04 Predictive Modeling

- Trained **LSTM neural networks** and **Linear Regression** models
- Forecasted price movement based on sentiment / historical data
- Evaluated using **RMSE** and correlation metrics

TIMELINE

Key trends and events in the crypto market & Coinbase milestones

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2021 | 2022 | 2023 | 2024 | 2025

May 2021
China's Crypto Ban
- China banned all cryptocurrency transactions and mining
- significant market downturn

June 2022
Layoffs & Market Downturn
- Coinbase laid off 18% of its workforce
- declining sentiment & trading volume
- spike in negative Reddit posts

May 2023
International Exchange Launch
- global expansion → increased search interest & positive sentiment
- temporary stock rebound

November 2024
Bitcoin Record High
- Bitcoin hit $95K following Trump's re-election
- Cross-correlation: led search interest by 3 days before price movement.

March 2025
Q1 COIN 30% drop
- COIN price dropped 30%
- negative sentiment spikes on news / media

STATISTIC 4

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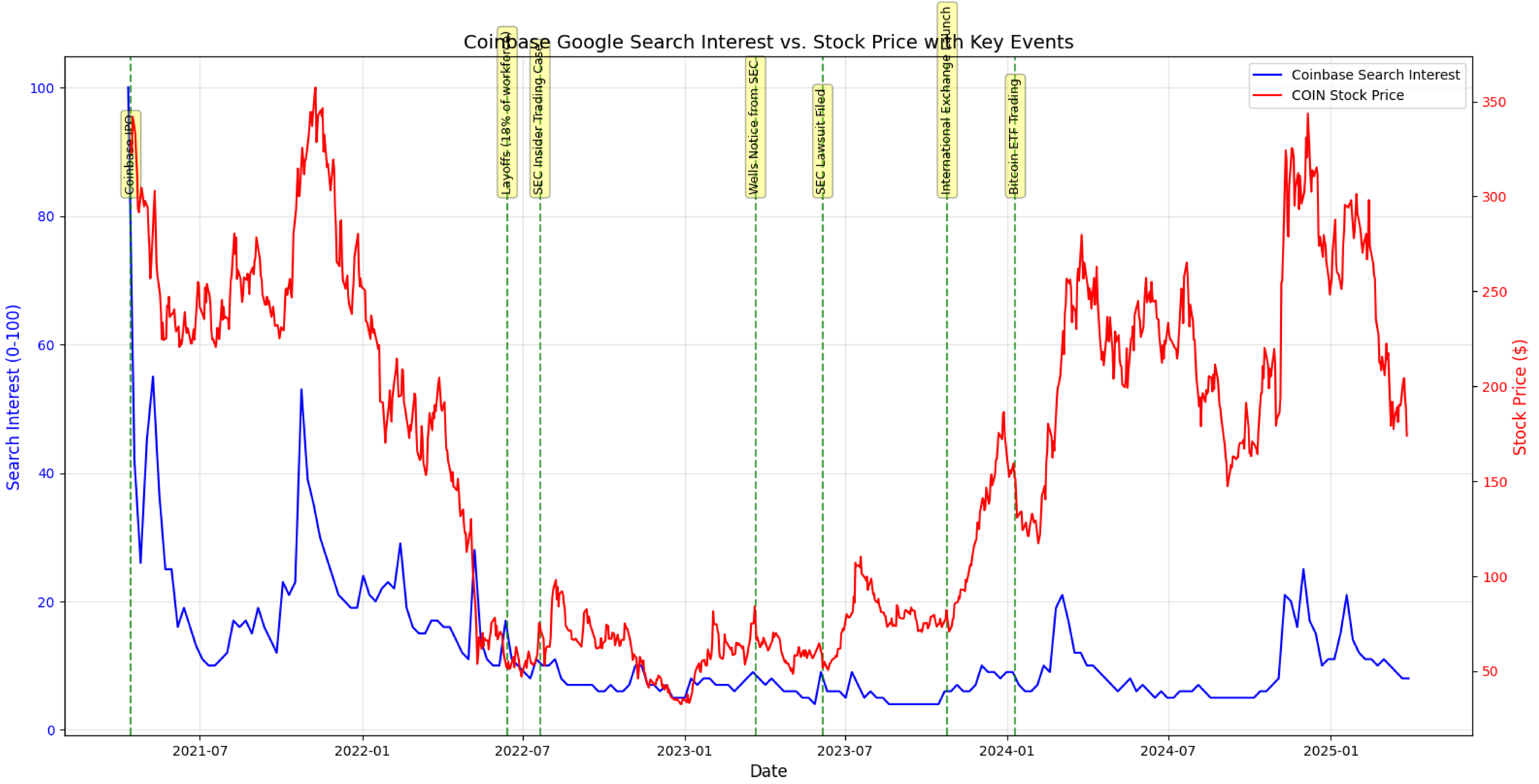

STATISTIC 5

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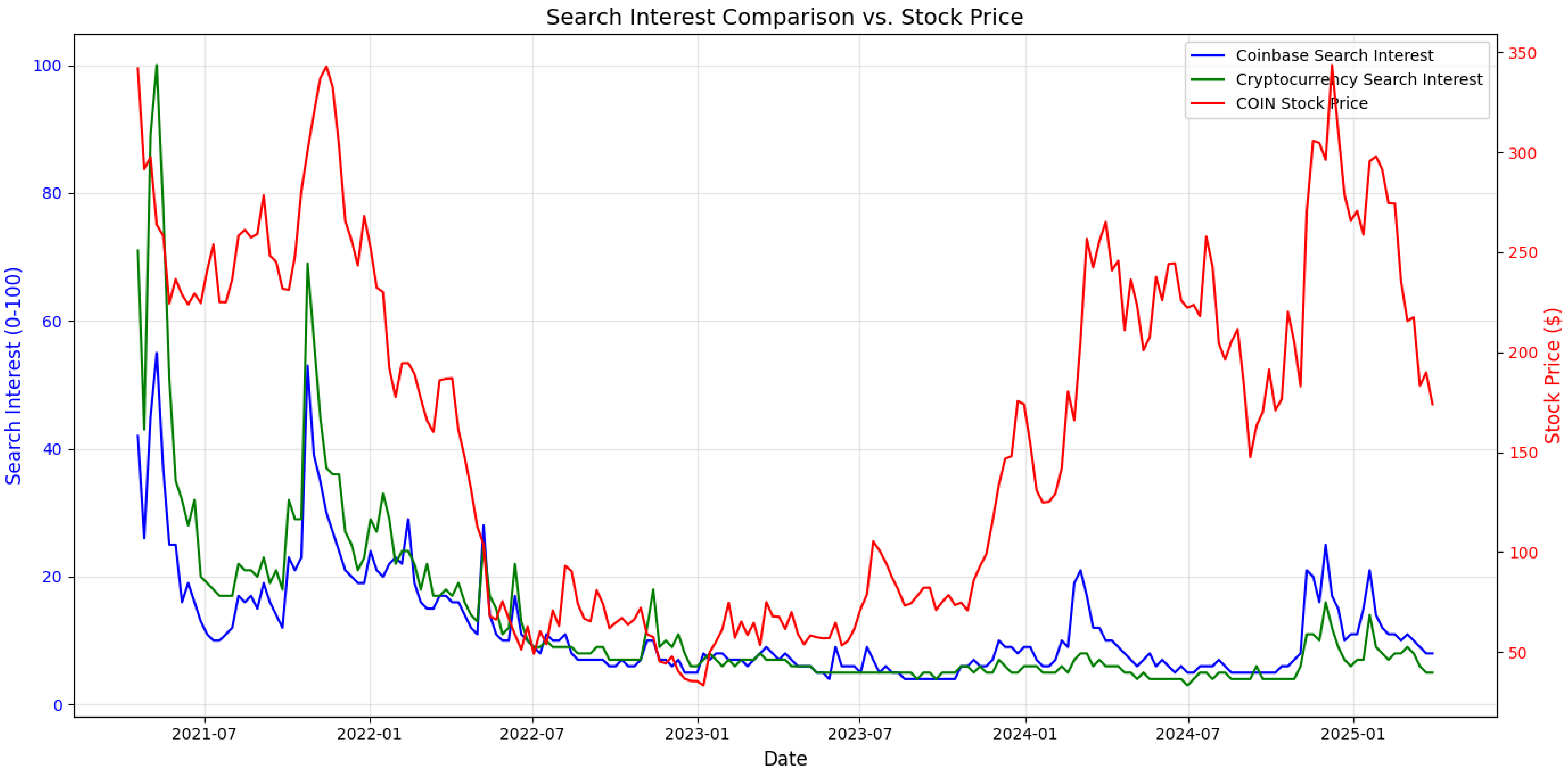

STATISTIC 1

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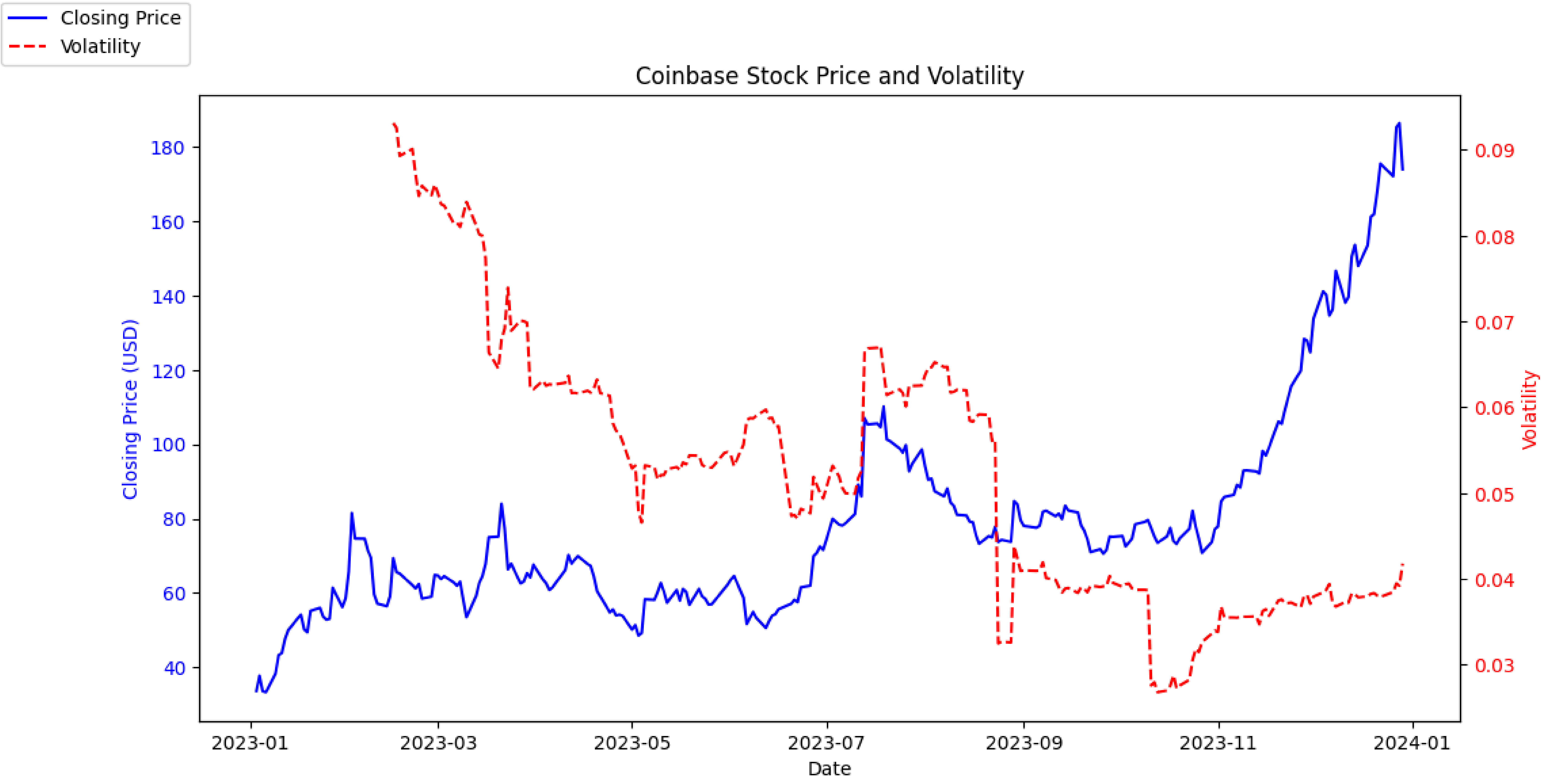

FAILED ITERATIONS

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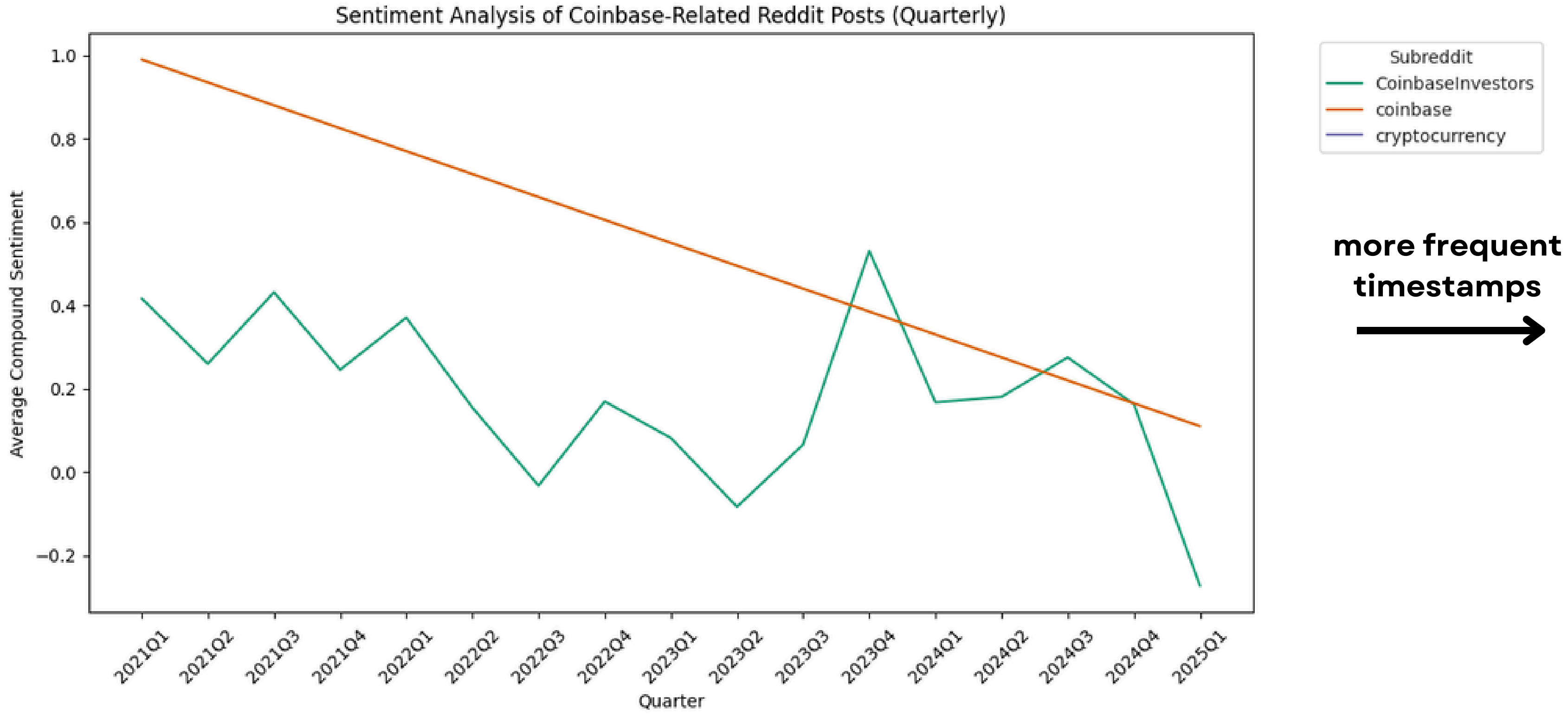
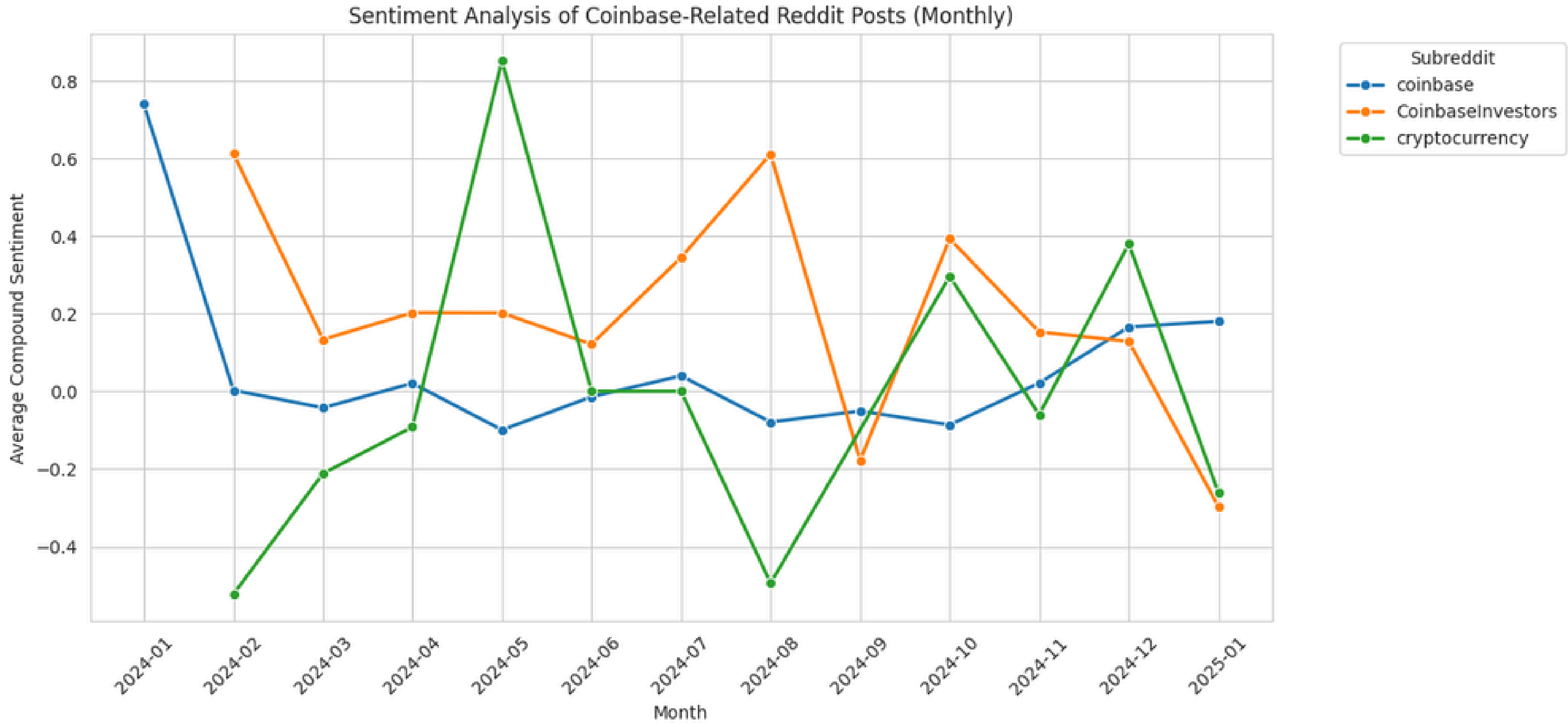


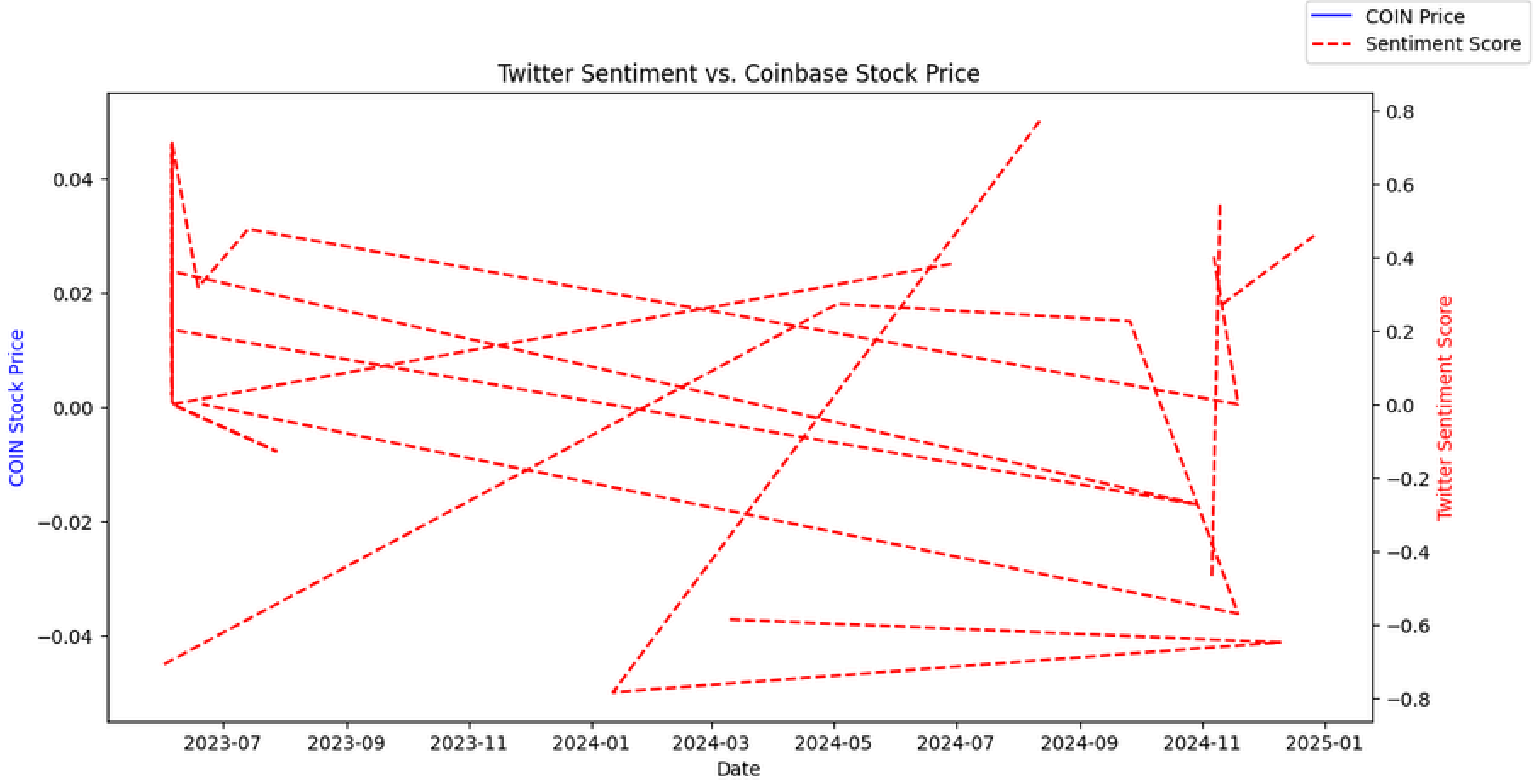
more frequent timestamps →




```
YFRateLimitError: Too Many Requests. Rate limited. Try after a while.
```

14	Political Events	Cointelegraph@CointelegraphâNov 7, 2024 BULLIS...	2024-11-07	2024.0	0.4019	0.144	0.000	0.856
15	Political Events	*Walter Bloomberg@DeItaoneâJan 21CRYPTO HAS A ...	NaT	NaN	0.0000	0.000	0.000	1.000
16	Political Events	Alymbis@alymbis2âJan 29 Coinbase doubles down ...	NaT	NaN	0.0000	0.000	0.000	1.000
17	Political Events	eimanbadwy@eimanbadwy2110âJan 29Coinbase's str...	NaT	NaN	0.5256	0.086	0.000	0.914

```
Extracted subtitles for 0 videos.
```



STATISTIC 2 + 3

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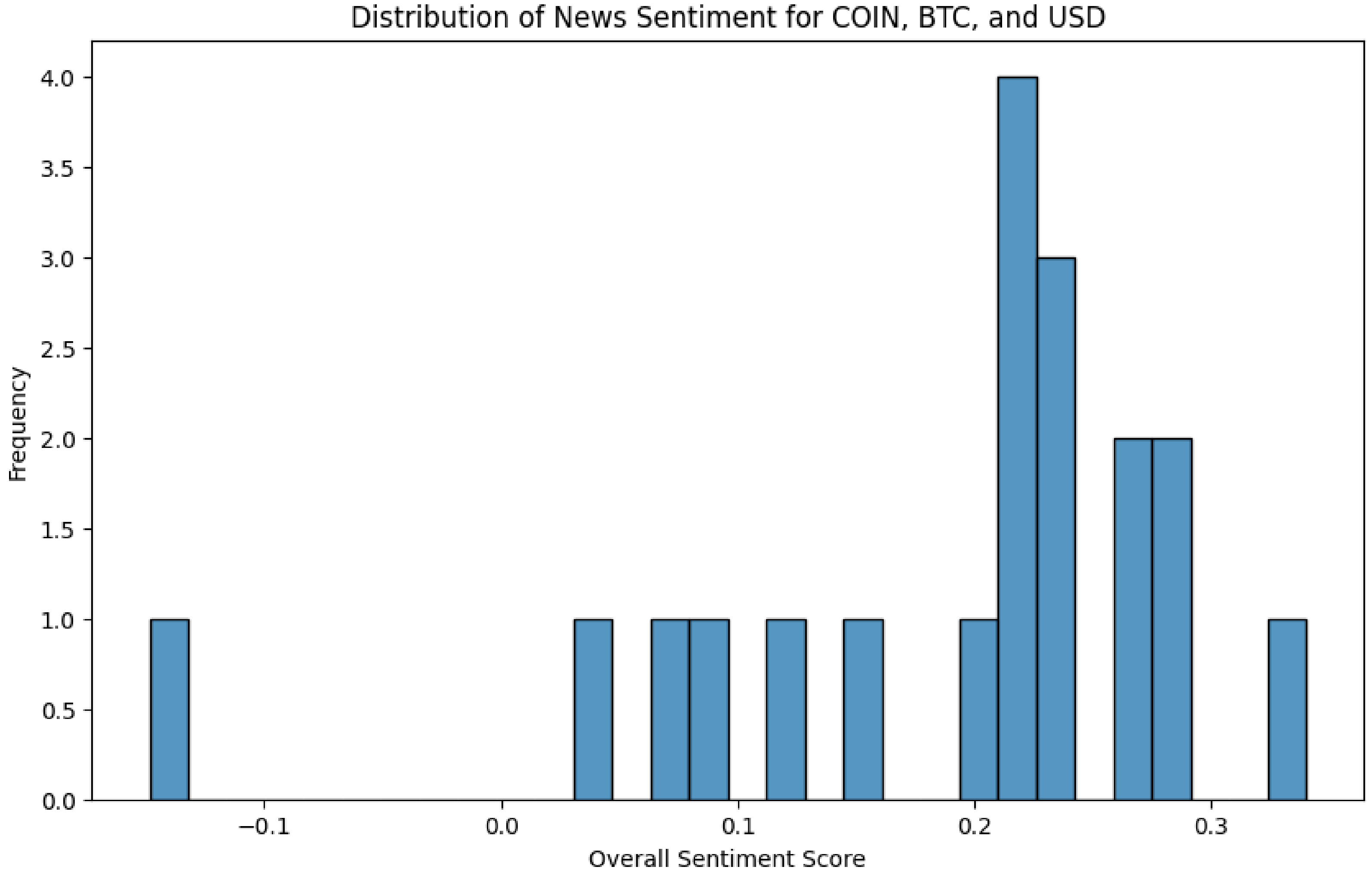

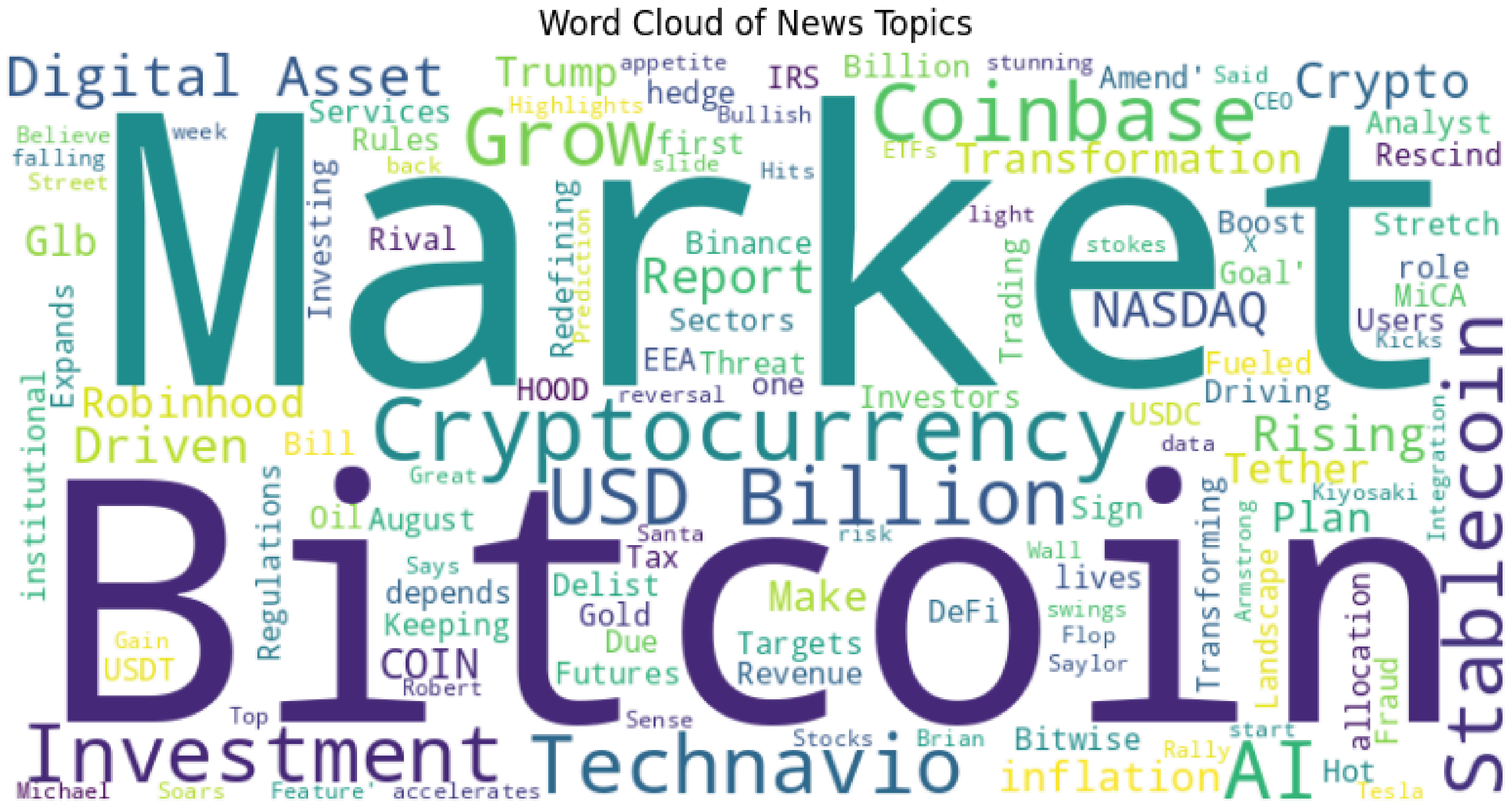

News Sentiment

- Overall neutral - positive sentiment
- Too complex to see clear sentiment
- Small sample size

News Topics Word Cloud

- The public generally associates the overall market with Bitcoin
- Sentiment of Bitcoin (as well as Coinbase, Stablecoin to a lesser extent) drives market

CODE SNIPPET

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```
# Calculating correlation between Google Trends data and stock price

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import yfinance as yf

# Get Coinbase stock data for the same period as our Google Trends data
start_date = trends_data['Week'].min().strftime('%Y-%m-%d')
end_date = trends_data['Week'].max().strftime('%Y-%m-%d')

print(f"Fetching Coinbase stock data from {start_date} to {end_date}...")
coin_stock = yf.download("COIN", start=start_date, end=end_date)

combined_df = pd.DataFrame()

# setting the stock data to weekly frequency to match Google Trends
weekly_stock = coin_stock['Close'].resample('W-FRI').last()

# create a DataFrame with both datasets
combined_df['Search_Interest'] = trends_data.set_index('Week')['Interest']
combined_df['Stock_Price'] = weekly_stock

combined_df = combined_df.dropna()

correlation = combined_df['Search_Interest'].corr(combined_df['Stock_Price'])
print(f"Correlation between Google search interest and COIN stock price: {correlation:.4f}")

if abs(correlation) < 0.3:
    interpretation = "weak or no linear relationship"
elif abs(correlation) < 0.5:
    interpretation = "moderate linear relationship"
else:
    interpretation = "strong linear relationship"

direction = "positive" if correlation > 0 else "negative"
if abs(correlation) >= 0.3:
    relationship = f"{direction} {interpretation}"
    print(f"This indicates a {relationship} between search interest and stock price.")
```

set timeframe

webscraping through yahoo finance as yf

merged dataframes

correlation analysis

NLP ANALYSIS

1st Attempt: LDA + VADER

- Topic Modeling: LDA
  - Identifies topics in Reddit posts by clustering word distributions
  - Helps understand *what* people are talking about
- Sentiment Classifier: VADER
  - Rule-based, uses dictionary of words with sentiment weights
  - Misclassifies nuanced or sarcastic text

LDA topics were vague due to short Reddit titles. VADER mislabeled strongly negative posts as positive.

Result: Topics were detected, but the sentiment attached to them was unreliable.

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1st Iteration - Reddit Sentiment w/ LDA + VADER

```
from nltk.sentiment import SentimentIntensityAnalyzer
nltk.download('vader_lexicon')
lda = LatentDirichletAllocation(n_components=num_topics, random_state=42)
```

	subreddit	title	upvotes	url	date	year	compound_sentiment	positive	negative	neutral
1028	coinbase	My CB account hacked after 10 years...	265	https://www.reddit.com/r/CoinBase/comments/1ht...	2025-01-04 20:32:58	2025	0.9949	0.198	0.046	0.756

r/CoinBase • 3 mo. ago
jtocontent

My CB account hacked after 10 years...

Discussion

The day after Christmas, I got two emails from Coinbase letting me know there had been withdrawals from my account —XRP and Solana, worth over $20K. I assumed they were phishing scams because, honestly, who trusts emails like that? So I deleted them without even opening them.

But something didn't sit right. I logged into my Coinbase account, and sure enough, the emails were legit. The funds were gone. Just... gone. I froze my account immediately, only to realize that freezing it also froze my ability to reach out to Coinbase support. Fantastic system design.

The weirdest part? My Bitcoin—much more valuable than the XRP and Solana—was untouched. It's like the hacker had some kind of moral code: "*I'll take the altcoins, but the BTC stays.*" Naturally, I moved all of it into cold storage immediately.

When I finally managed to connect with Coinbase support through their chat system, the first response was a classic: "*Once the funds are transferred, there's nothing we can do.*" Great. But after an hour of painfully slow back-and-forth, the agent gave me a faint glimmer of hope: "*There's a slim chance you might recover your funds... someday... maybe.*"

Unsatisfied, I pulled some strings and spoke with an actual person—a second cousin of a friend who works at Coinbase customer support. Surely a real human would offer something better. His advice? "*Move whatever you have left to cold storage and accept that your XRP and Solana are probably gone forever.*"

On a 2nd chat with CB support I was informed I wasn't the only one this had happened to and that CB was looking into the issue and would get back to me... told me to check my email in a week or so. I've screenshot both chats as proof.

NLP ANALYSIS

2nd Attempt: BERTopic + VADER

- Topic Modeling: BERTopic
  - Uses transformer-based embeddings + clustering
  - Better suited for short-form text like Reddit titles
- Sentiment Classifier: VADER
  - Rule-based, uses dictionary of words with sentiment weights
  - Continues to misclassify sarcasm and emotional nuance

Hypothesis:
LDA wasn't modeling topics well, maybe that was causing bad sentiment outputs."
Reality:
BERTopic improved topic quality, but sentiment errors remained.

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2nd Iteration - Reddit Sentiment w/ BERTopic + VADER

```
from nltk.sentiment import SentimentIntensityAnalyzer

nltk.download('vader_lexicon')

topic_model = BERTopic(language="english")
```

	subreddit	title	upvotes	url	date	year	compound_sentiment	positive	negative	neutral
0	coinbase	I got scammed out of 147,592$ worth of ETH	1302	https://www.reddit.com/r/CoinBase/comments/1hf...	2024-12-16 12:35:36	2024	0.8389	0.114	0.085	0.801
1	coinbase	See Trump coin on Coinbase turns my stomach.	1093	https://www.reddit.com/r/CoinBase/comments/1i6...	2025-01-21 22:31:18	2025	0.8720	0.230	0.034	0.736
2	coinbase	Got scammed on Coinbase and lost 41 ETH ($166k!)	1067	https://www.reddit.com/r/CoinBase/comments/1hd...	2024-12-13 20:09:44	2024	-0.9928	0.084	0.114	0.803

2nd Iteration - The issue was not what people were talking about (topics), but how the model understood their tone (sentiment). The real fix would require replacing VADER, not LDA.

NLP ANALYSIS

3rd Attempt: BERTopic + RoBERTa

- **Topic Modeling: BERTopic**
  - **Still used to extract discussion themes from Reddit posts**
- **Sentiment Classifier: RoBERTa**
  - **A transformer-based model trained on Twitter data**
  - **Capable of understanding context, sarcasm, and informal tone**

- **Negative sentiment dominates in most months, especially in May and November.**
- November stands out with the highest volume of posts overall, and a spike in both negative and neutral sentiment..

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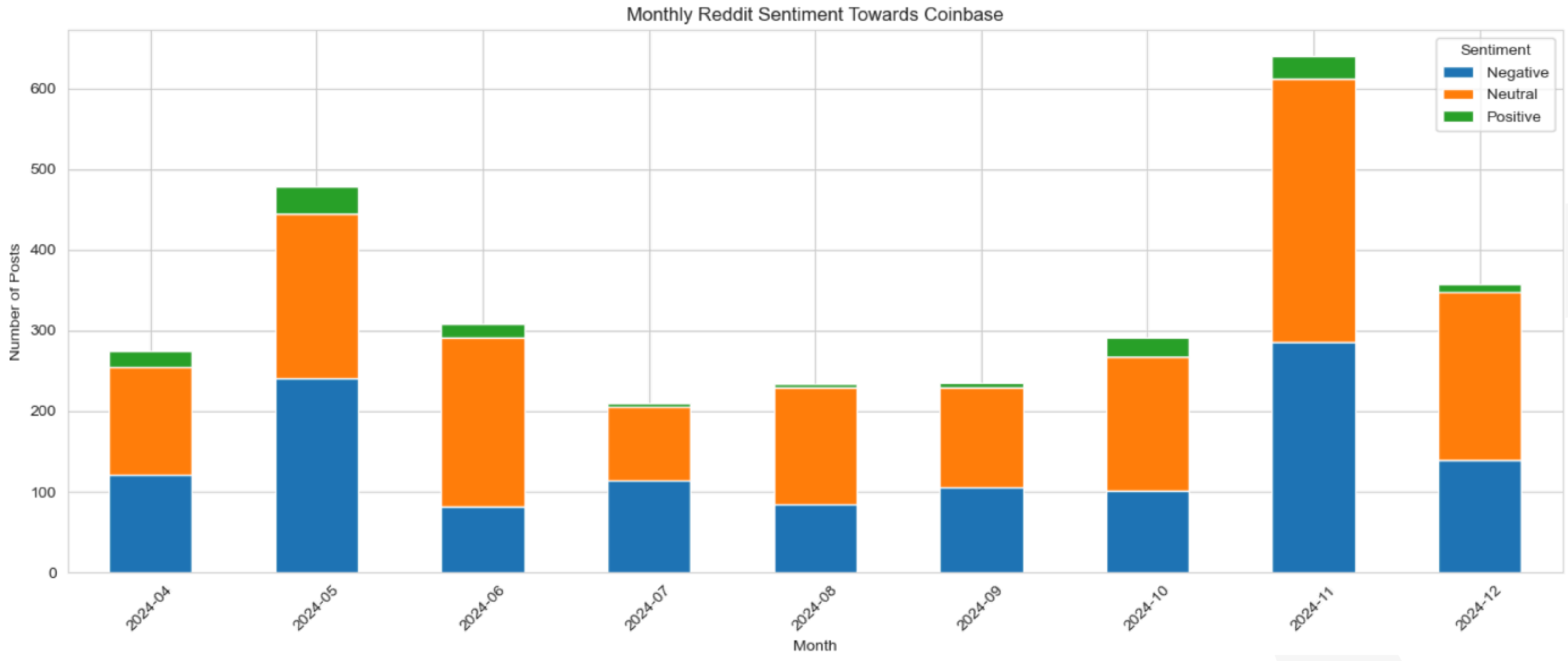
3rd Attempt – RoBERTa for Accurate Sentiment Classification

```
from transformers import pipeline
sentiment_pipeline = pipeline("sentiment-analysis", model="cardiffnlp/twitter-roberta-base-sentiment")
df["RoBERTa_Sentiment"] = df["title"].apply(classify_sentiment)
```

	title	RoBERTa_Sentiment
0	I got scammed out of 147,592$ worth of ETH	Negative
1	Got scammed on Coinbase and lost 41 ETH ($166k!)	Negative
2	Let's Bring $PEP to Coinbase! 🚀	Positive
3	I STRONGLY suggest that you get rid of your as...	Negative
4	Coinbase pretty much stole my btc.	Negative



REDDIT SENTIMENT

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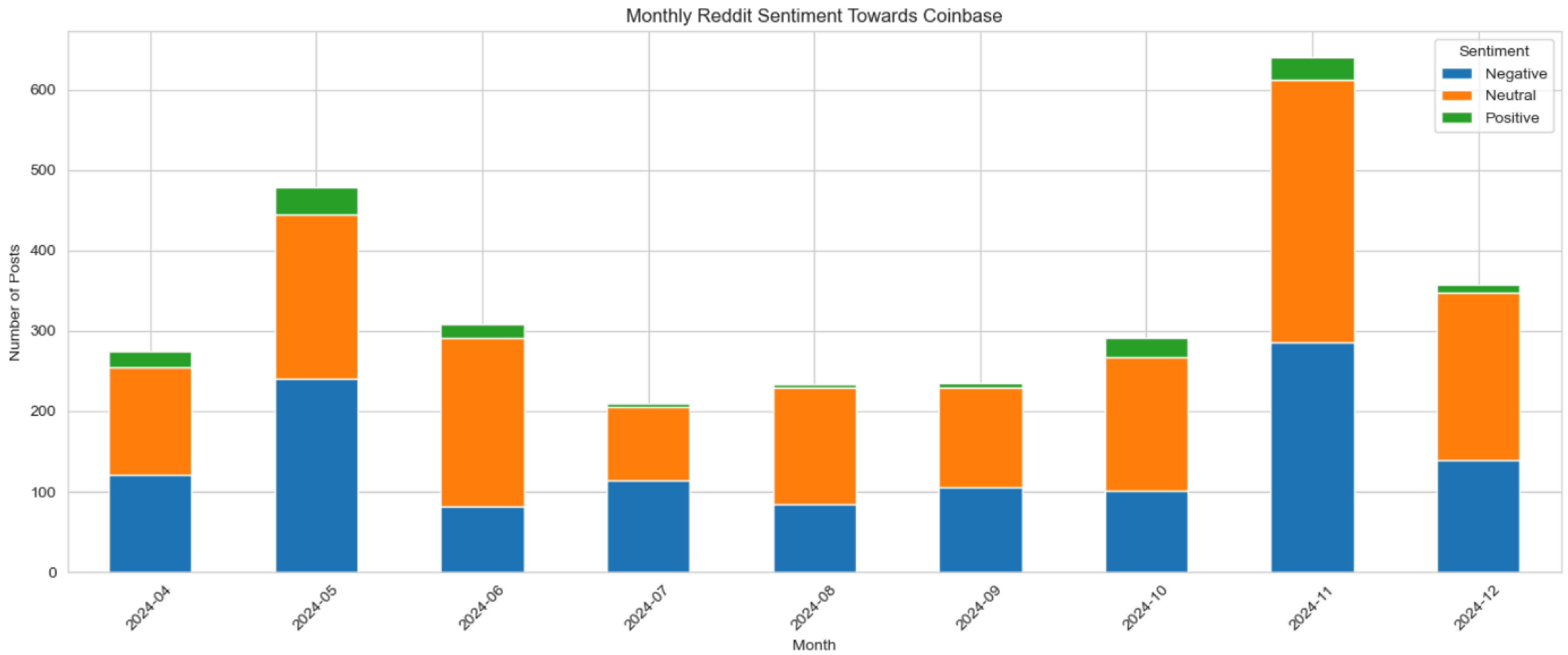
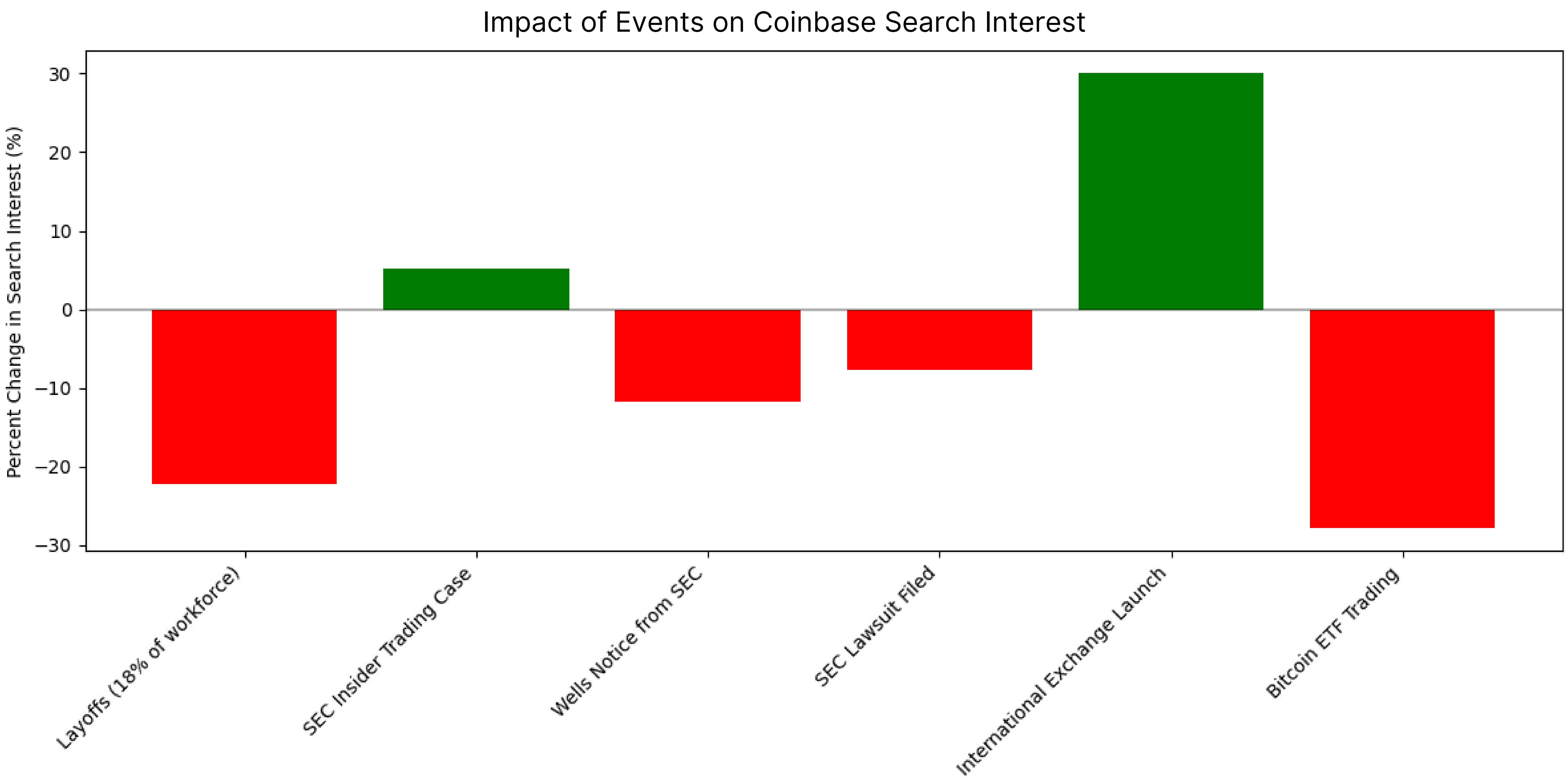


May 2024:
Event: On May 2, Coinbase released its Q1 2024 financial results, reporting significant profits due to a surge in crypto prices.
Stock Performance: COIN's stock closed at $225.92 on May 31, marking a 13.5% increase for the month.
Sentiment Analysis: Despite positive financial news, Reddit sentiment remained predominantly negative, possibly reflecting user concerns over other issues not directly related to financial performance.

November 2024:
Regulatory Challenges: In July, Coinbase's UK unit was fined for violating financial crime regulations, leading to a class-action lawsuit filed by investors with a deadline in November.
Political Climate: The U.S. presidential election in early November resulted in a Republican victory, leading to expectations of lighter crypto regulations.
Stock Performance: COIN's stock experienced significant volatility, with a high of $341.75 and closing at $296.20 by the end of November.
Sentiment Analysis: Reddit sentiment was mixed, with spikes in both negative and neutral sentiments, reflecting the regulatory uncertainties and political developments.

STATISTIC 6



Impact of Events on Coinbase Search Interest
Percent Change in Search Interest (%)
30
20
10
0
−10
−20
−30
Layoffs (18% of workforce)
SEC Insider Trading Case
Wells Notice from SEC
SEC Lawsuit Filed
International Exchange Launch
Bitcoin ETF Trading

SEARCH INSIGHTS

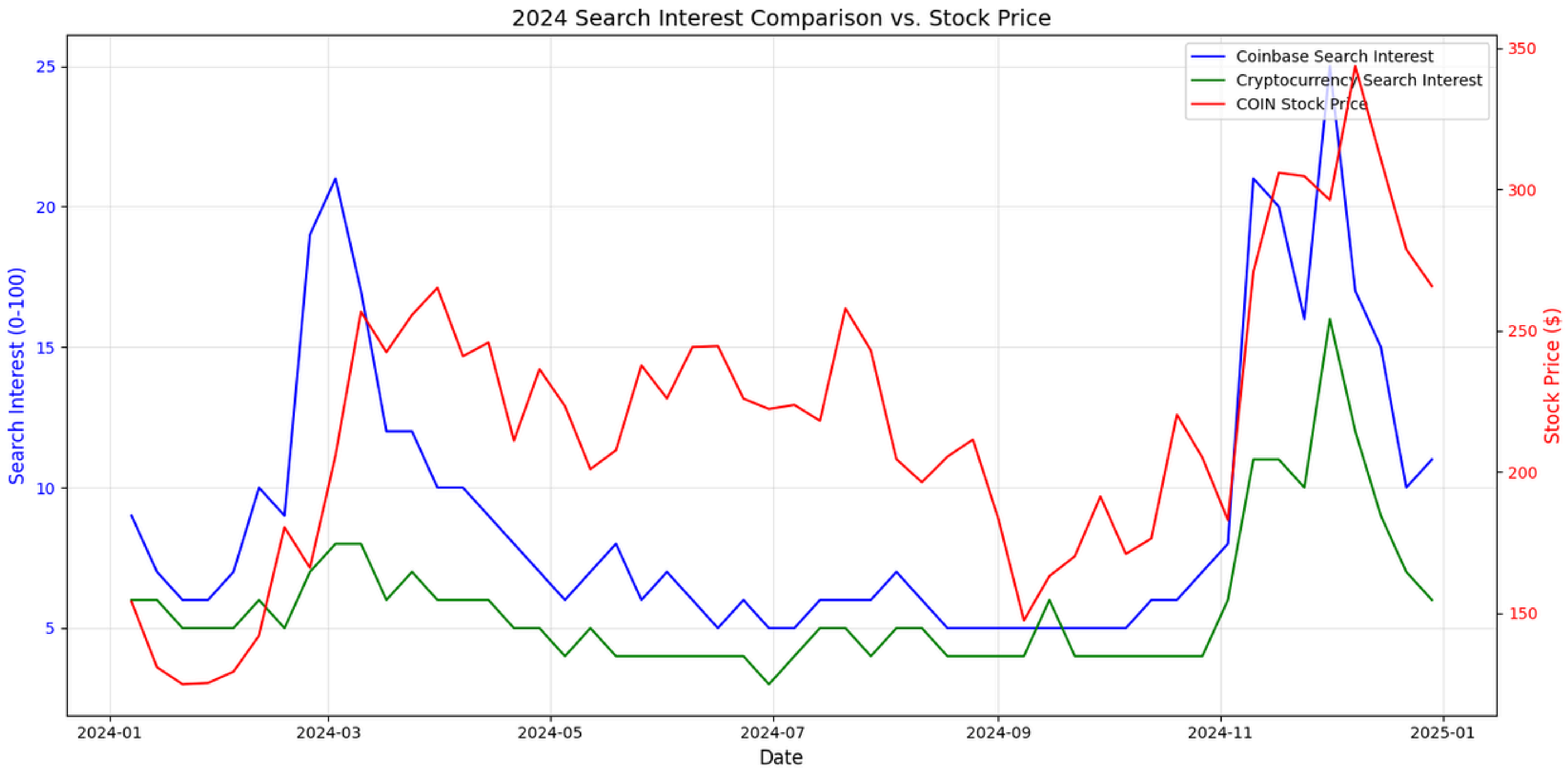
Coinbase and cryptocurrency search interest tends to typically spike **before** stock price moves

In 2024, these spikes acted as early warning signals ahead of major volatility.

Google Trends is a *valuable leading indicator* for investor attention and market sentiment.

Search Interest vs COIN Stock Price (2024)

Spikes in search activity often precede price movement



In 2024, we observed that spikes in Google search interest for "Coinbase" and "cryptocurrency" often preceded major price movements.

MODELS

The LSTM Model uses these features:

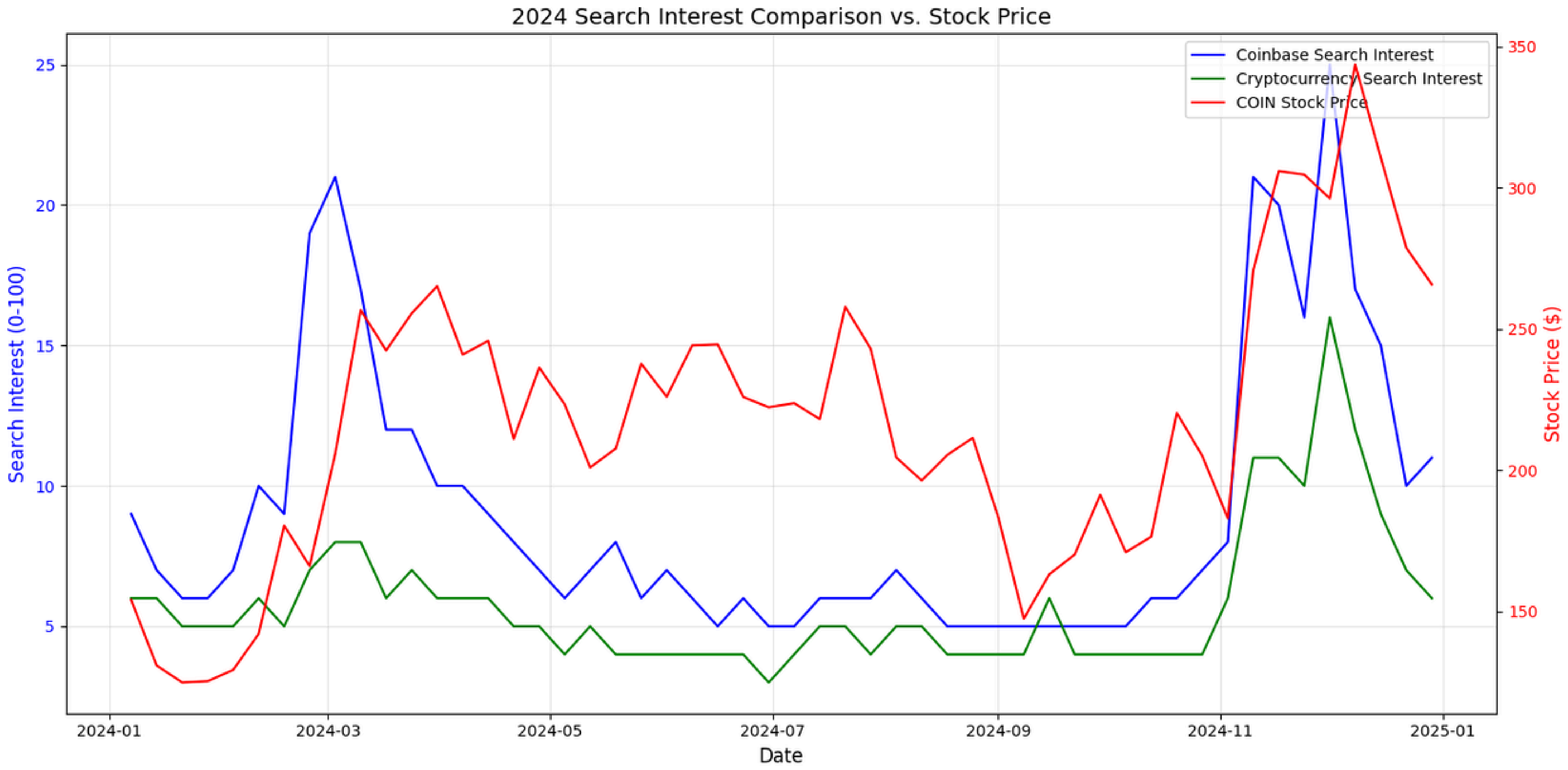
1. Coinbase search interest
2. Cryptocurrency search interest
3. Historical COIN stock prices

(features normalized + lagged)

The Linear Regression model outperforms the LSTM model which is trained on time-series data.

Predicted vs. Actual COIN Stock Prices – LSTM vs Linear Regression

(2024–2025 test set)



LSTM

Correlation between predicted and actual prices: 0.7731

Linear Regression

Correlation between predicted and actual prices: 0.8604

ML FORECASTING

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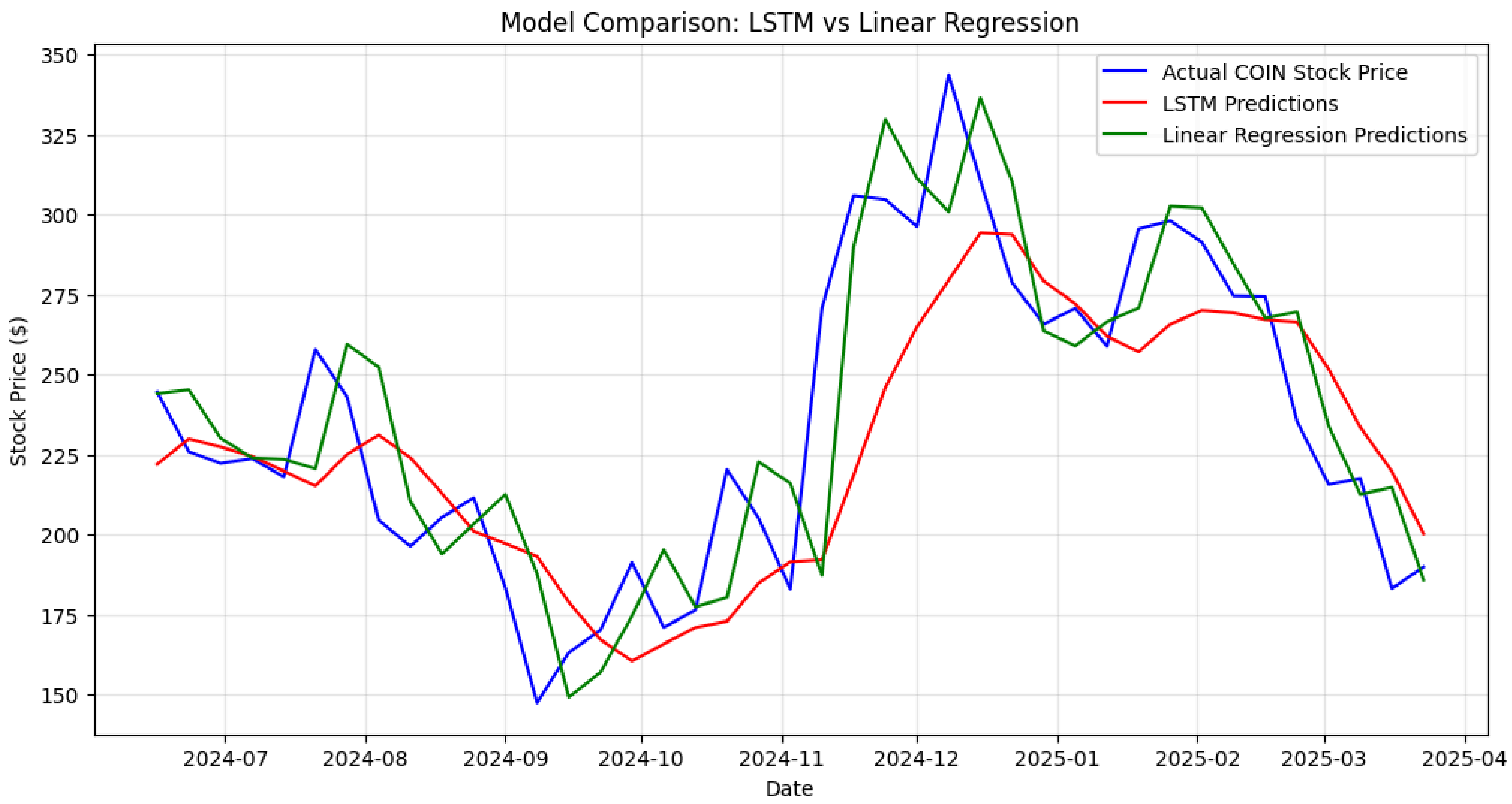

Both models captured major trends and price swings.

Linear Regression followed price direction more closely than LSTM, though LSTM smoothed noise better which is useful for longer term forecasting.

STATISTIC 7

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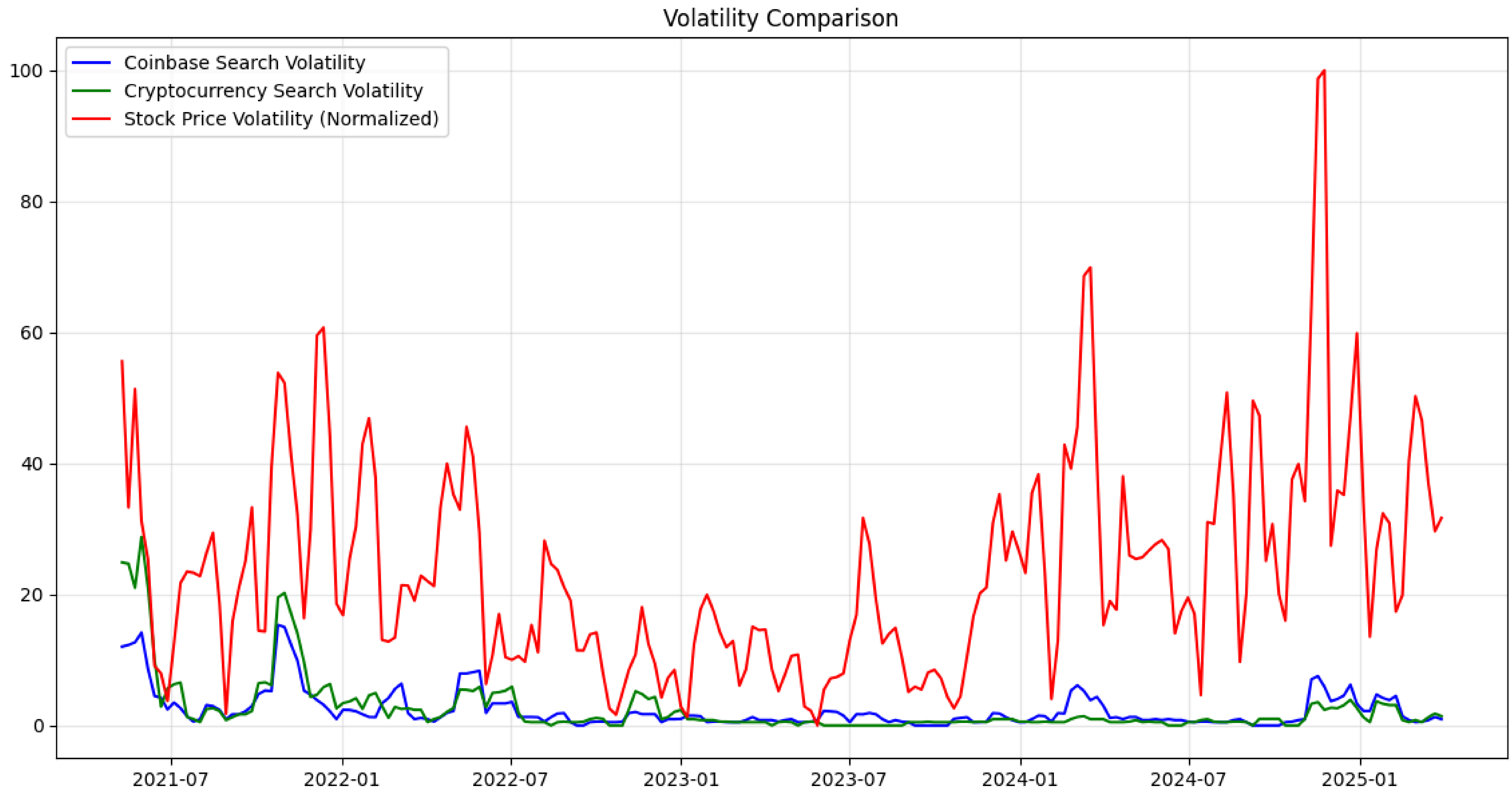

CODE SNIPPET

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LSTM Model

```
model = Sequential()

# LSTM layer with 50 units
model.add(LSTM(50, return_sequences=True, input_shape=(X_train.shape[1], X_train.shape[2])))
model.add(Dropout(0.2))  # Dropout to prevent overfitting

# second LSTM layer
model.add(LSTM(50, return_sequences=False))
model.add(Dropout(0.2))

# dense output layer
model.add(Dense(1))

# compile the model
model.compile(optimizer=Adam(learning_rate=0.001), loss='mean_squared_error')

# model summary
model.summary()

# train the model
print("Training the LSTM model...")
history = model.fit(
    X_train, y_train,
    epochs=50,
    batch_size=4,
    validation_split=0.2,
    verbose=1
)
```

Model: "sequential"

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 4, 50)	10,800
dropout (Dropout)	(None, 4, 50)	0
lstm_1 (LSTM)	(None, 50)	20,200
dropout_1 (Dropout)	(None, 50)	0
dense (Dense)	(None, 1)	51

Total params: 31,051 (121.29 KB)
Trainable params: 31,051 (121.29 KB)
Non-trainable params: 0 (0.00 B)

Linear Regression

```
# now going to implement a simple linear regression for comparison
if 'X_train' in locals() and len(X_train) > 0:
    from sklearn.linear_model import LinearRegression

    # reshaping data for linear regression
    X_train_flat = X_train.reshape(X_train.shape[0], X_train.shape[1] * X_train.shape[2])
    X_test_flat = X_test.reshape(X_test.shape[0], X_test.shape[1] * X_test.shape[2])

    # training linear regression model
    print("Training a simple Linear Regression model for comparison...")
    lr_model = LinearRegression()
    lr_model.fit(X_train_flat, y_train)

    lr_predictions = lr_model.predict(X_test_flat)

    lr_predicted_prices = price_scaler.inverse_transform(lr_predictions.reshape(-1, 1))
```

!! failed ARIMA model

```
# Perform time series decomposition
try:
    # Try weekly seasonality first
    decomposition = seasonal_decompose(ts_data, model='multiplicative', period=4)

    # Plot decomposition
    fig, (ax1, ax2, ax3, ax4) = plt.subplots(4, 1, figsize=(12, 10))
    decomposition.observed.plot(ax=ax1)
    ax1.set_title('Observed')
    decomposition.trend.plot(ax=ax2)
    ax2.set_title('Trend')
    decomposition.seasonal.plot(ax=ax3)
    ax3.set_title('Seasonality')
    decomposition.resid.plot(ax=ax4)
    ax4.set_title('Residuals')
    plt.tight_layout()
    plt.show()

    # ARIMA forecasting
    # Split data for training and testing
    train_size = int(len(ts_data) * 0.8)
    train, test = ts_data[:train_size], ts_data[train_size:]

    # Fit ARIMA model
    model = ARIMA(train, order=(1, 1, 1))
    model_fit = model.fit()

    # Forecast
    forecast = model_fit.forecast(steps=len(test))
```

FAILED ITERATIONS

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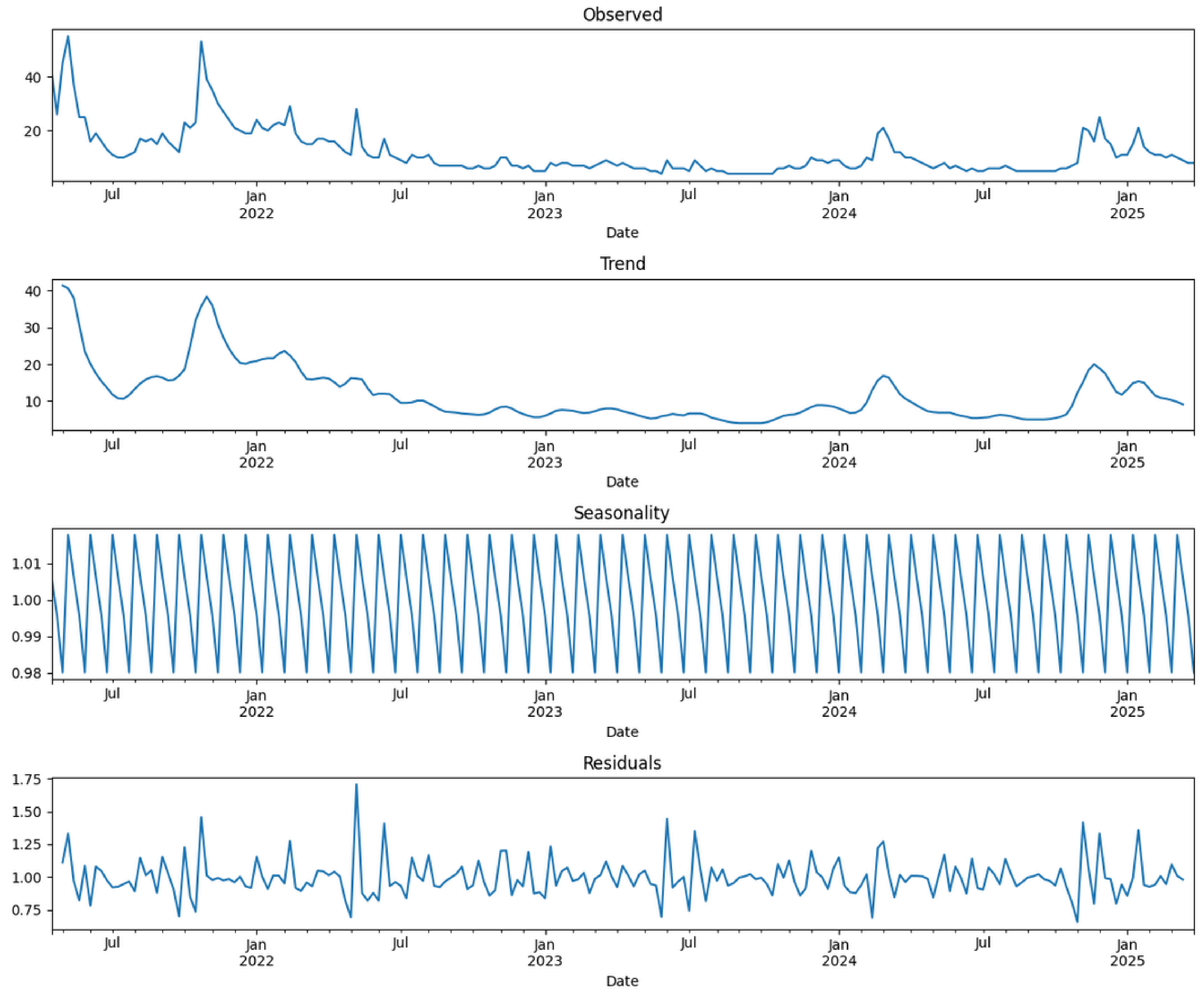

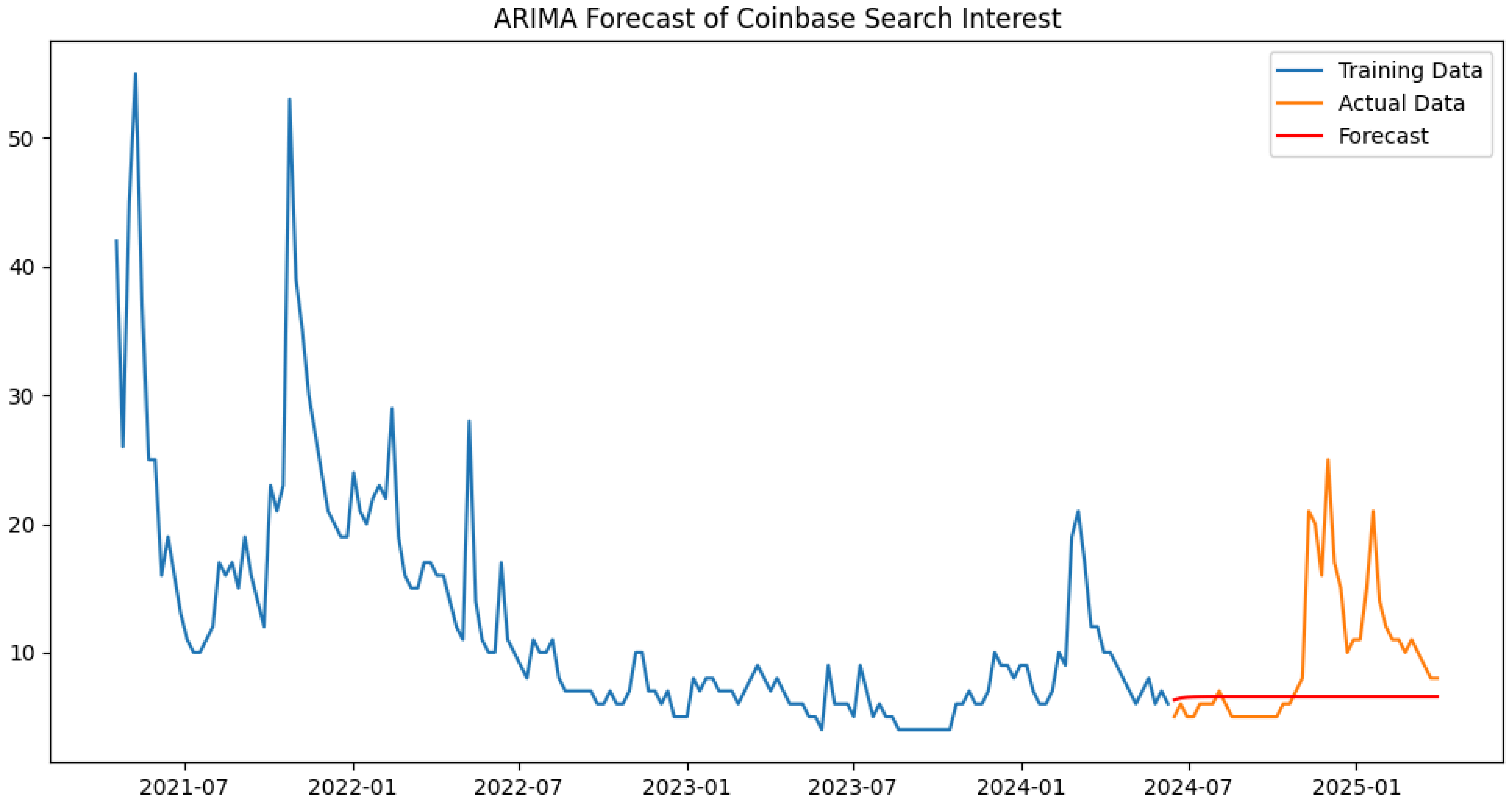


```
Forecast Accuracy Metrics:
Root Mean Squared Error: 6.05
Mean Absolute Error: 4.07
```

We also tested an ARIMA model, but it underperformed with an RMSE of 6.05 - probably due to the model's limitations on sparse sentiment features.

STRATEGY

- leverage volatile market
  → gain profit
- associated risks
- our predictions

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01 Trading Opportunities

- crypto stocks (eg COIN) move with bitcoin prices & same volatility
- Track sentiment & overall activity for leading signals
- Event-driven investing might outperform statistical predictions.

02 Risk Management

- Sentiment can change quickly - therefore false positives are possible, hard to time entry/exit
- Overreliance on search trends may miss structural risks
- API limitations and sparse data reduce real-time accuracy

03 Future Forecasting

- Sentiment volatility is most likely to stay predictive
- More granular data → realtime price forecasting *could* become viable (could also try reinforcement learning)
- Our framework could be adapted to other crypto-related equities

CONCLUSION

- Sentiment = strong leading indicator of price
- Volatility in search interest precedes stock price swings
- ML models can capture directional trends, but linear models performed surprisingly well

01 Overall Trends

- Public sentiment consistently leads crypto asset price moves
- Volatility in search interest predicts major stock swings 3-5 days ahead
- Simple models (linear regression) outperformed complex ML models for noisy crypto data

02 Limitations

- API rate limits and restricted access to real-time data
- Reddit and X timestamps were too coarse for intraday prediction
- Not enough data (ARIMA ineffective)

03 Future Steps / Predictions

- Build real-time crypto sentiment pipelines
- Expand coverage beyond $COIN to Bitcoin (BTC), Ethereum (ETH), and DeFi assets
- Integrate alternative data sources (eg. Bitcoin wallet activity) to strengthen signals
- Quantify “sentiment shocks” as an early warning system

TAKEAWAYS

Learnings

- Google search interest consistently leads price movements by 3-5 days.
- Public sentiment is a stronger predictor for crypto stocks than traditional equities.
- Simple linear regression models outperformed complex LSTM networks for crypto price forecasting.
- Reddit provided stronger sentiment signals than X .

Things we wish we did differently

- Develop custom crypto-specific sentiment dictionaries for improved accuracy
- Webscrape through more / diffferent APIs, including premium or official endpoints

Challenges faced

- API rate limitations required developing custom data collection pipelines
- Social media signal-to-noise ratio requires us to use more advanced filtering techniques

Actionable Next Steps

- Develop a real time sentiment dashboard for trading desk integration
- Expand analysis to broader crypto asset ecosystem beyond Coinbase (BTC, ETH, DeFi tokens)

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THANK YOU

FOR YOUR ATTENTION

April 11, 2025

Special thanks to our mentor:

Elif Babaoğlu